

NEAR-WHOLE GENOME SEQUENCING OF SARS-CoV-2 USING POOLED AMPLICONS ON ILLUMINA PLATFORMS

Part 1 - Viral RNA extraction

Here we provide an example low throughput viral RNA extraction protocol; however, any platform that extracts viral or pathogen RNA (or total nucleic acid) should be suitable. We typically use the residual viral extracts from clinical samples following diagnostic RT-PCRs, which include extracts off both Qiagen BioRobot EZ1 and Roche MagnaPure 96 platforms. Please ensure appropriate PPE and containment to avoid exposure to infectious samples, noting that following the addition of the DNA/RNA Shield, the virus will be inactivated.

1. Combine 100 µl 2X DNA/RNA Shield with 100 µl respiratory sample (sputum, aspirate or swab medium), then mix well by gentle vortexing.
2. Add 400 µl Viral RNA Buffer to each 200 µl sample and mix well by gentle vortexing.
3. Transfer the mixture into a Zymo-Spin IC column placed in a collection tube. Centrifuge at 12,000g for 2 mins to bind viral RNA to matrix.
4. Transfer the column into a new collection tube.
5. Add 500 µl Viral Wash Buffer to the column, centrifuge at 12,000g for 30 sec.
6. Transfer the column into a new collection tube.
7. Repeat steps 5 & 6.
8. Add 500 µl ethanol (95-100%) to the column and centrifuge at 12,000g for 2 min to ensure complete removal of the wash buffer.
9. Transfer the column into a clean elution tube (2 ml Eppendorf PCR clean, DNA LoBind tube).
10. Add 25 µl DNase/RNase-Free Water directly to the column matrix, incubate for 5 mins and then centrifuge at 12,000g for 30 sec to elute the viral RNA.
11. The eluted viral RNA can be used immediately or stored frozen at -80°C.

Part 2 - cDNA synthesis

cDNA is prepared from viral RNA using SuperScript IV VILO Master Mix, which combines both random and oligo-dT priming for first strand synthesis. The oligo-dT priming is essential as the reverse primer for the final amplicon sits close to the viral poly-A tail in the 3' UTR.

1. Setup the following reaction mix for each sample and a no template control (NTC):

REAGENT	VOLUME (µl)
SuperScript IV VILO MasterMix (5X)	5
Nuclease free water	15
Viral RNA	5
<i>TOTAL</i>	25

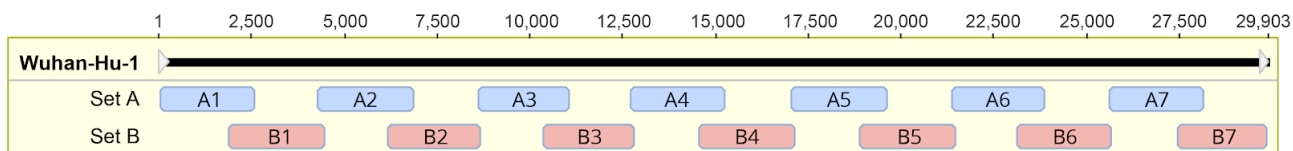
2. Incubate the reaction as follows:

STEP	TEMP (°C)	TIME (mm:ss)
Priming	25	10:00
Extension	50	20:00
Denature RT	85	05:00
Hold	4	∞

- Keep the cDNA on ice if being used immediately or store frozen at -20°C

Part 3 - Genome amplification

Next we take the cDNA prepared from viral RNA and amplify 14 regions (~2.5kb each) that tile across the viral genome. The PCRs are performed in parallel and is NOT a multiplexed reaction. The general approach is demonstrated by the following schematic:

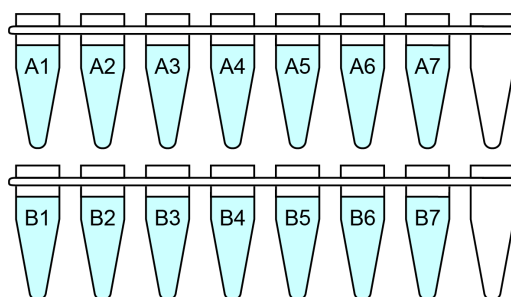


- The amplicon primers are as follows:

PCR	PRIMER	SEQUENCE (5' - 3')	LENGTH (nt)
A1	SARS2_A1F_31	ACCAACCAACTTTTCGATCTCTTGT	2562
	SARS2_A1R_2569	GCTTCAACAGCTTCACTAGTAGGT	
A2	SARS2_A2F_4295	ACAGTGCTTAAAAAGTGTAAGTGCC	2579
	SARS2_A2R_6847	ACAGTATTCTTTGCTATAGTAGTCGGC	
A3	SARS2_A3F_8596	ACTTGTGTTCTTTTGTGCTGC	2479
	SARS2_A3R_11049	GAACAAAGACCATTGAGTACTCTGGA	
A4	SARS2_A4F_12711	TACGACAGATGTCTTGCTGC	2536
	SARS2_A4R_15225	TAACATGTTGTGCCAACCACCA	
A5	SARS2_A5F_16847	ACTATGGTGATGCTGTTGTTTACCG	2432
	SARS2_A5R_19254	ACCAGGCAAGTTAAGGTTAGATAGC	
A6	SARS2_A6F_21358	ACAAATCCAATTCAGTTGTCTTCCTATTC	2490
	SARS2_A6R_23823	TGTGTACAAAACTGCCATATTGCA	
A7	SARS2_A7F_25602	ACTAGCACTCTCCAAGGGTGTT	2571
	SARS2_A7R_28146	AGGTTCTTGCAATTAATTGTAAGG	
B1	SARS2_B1F_1876	ATCAGAGGCTGCTCGTGTTGTA	2575

	SARS2_B1R_4429	AGTTTCCACACAGACAGGCATT	
B2	SARS2_B2F_6287	TGGTGTATACGTTGTCTTTGGAGC	2565
	SARS2_B2R_8828	CACTTCTCTTGTTATGACTGCAGC	
B3	SARS2_B3F_10363	TGTTTCGCATTCAACCAGGACAG	2440
	SARS2_B3R_12780	CCTACCTCCCTTTGTTGTGTTGT	
B4	SARS2_B4F_14546	AGGAATTACTTGTGTATGCTGCTGA	2607
	SARS2_B4R_17131	ACACTATGCGAGCAGAAGGGTA	
B5	SARS2_B5F_18897	TGTTAAGCGTGTTGACTGGACT	2559
	SARS2_B5R_21428	TGACCTTCTTTTAAAGACATAACAGCAG	
B6	SARS2_B6F_23123	CCAGCAACTGTTTGTGGACCTA	2551
	SARS2_B6R_25647	AGGTGTGAGTAACTGTTACAAACAAC	
B7	SARS2_B7F_27447	TCACTACCAAGAGTGTGTTAGAGGT	2420
	SARS2_B7R_29837	TTCTCCTAAGAAGCTATTAAAATCACATGG	

2. Prepare the primers for each amplicon set by combining 5 μ l of forward primer (100 μ M), 5 μ l of reverse primer (100 μ M) and 190 μ l of nuclease free water into clean PCR strips, as below:



3. One standard RT-PCR reaction is shown below, but each sample and control requires 14 reactions, each containing the different amplicon primers A1-A7 & B1-B7

REAGENT	VOLUME (μ l)
Platinum SuperFi Green Mastermix (2X)	10
Nuclease free water	6.5
Primer pool (5 μ M)	2
Viral RNA	1.5
TOTAL	20

4. Setup the following reaction mix for each sample and the no template control (NTC) according to requirements (one sample has 15 reactions, 14 primers + 1 for pipetting error):

REAGENT	VOLUME (μ l), s=samples			
	S=1	S=3	S=6	S=12
Platinum SuperFi Green Mastermix (2X)	150	450	900	1800
Nuclease free water	97.5	292.5	585	1170

5. Dispense 16.5 μ l of the reaction mix into each well as required, a suggested plate setup is provided below for a 6 sample plate:

	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	NTC	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	NTC
	1	2	3	4	5	6	7	8	9	10	11	12
A	A1	A1	A1	A1	A1	A1	B1	B1	B1	B1	B1	B1
B	A2	A2	A2	A2	A2	A2	B2	B2	B2	B2	B2	B2
C	A3	A3	A3	A3	A3	A3	B3	B3	B3	B3	B3	B3
D	A4	A4	A4	A4	A4	A4	B4	B4	B4	B4	B4	B4
E	A5	A5	A5	A5	A5	A5	B5	B5	B5	B5	B5	B5
F	A6	A6	A6	A6	A6	A6	B6	B6	B6	B6	B6	B6
G	A7	A7	A7	A7	A7	A7	B7	B7	B7	B7	B7	B7
H	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty	Empty
Set A primers						Set B primers						

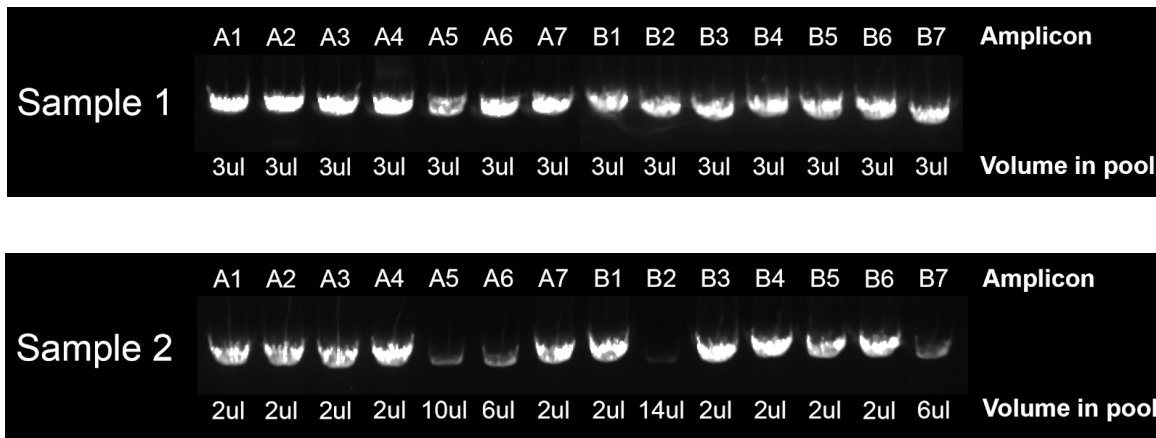
6. Using a 8-channel multipipettor, add 2 μ l of the diluted and premixed primer sets into each well according to the plate layout.
7. Using a 8-channel multipipettor, add 1.5 μ l of viral cDNA into each well according to the plate layout.
8. Seal the plate, and centrifuge for 30 sec in a plate spinner.
9. Incubate the reaction as follows:

STEP	TEMP ($^{\circ}$ C)	TIME (mm:ss)	CYCLES
Hot start	98	02:00	1X
Denaturation	98	00:15	40X
Annealing	65	00:30	
Extension	72	02:00	
Final extension	72	05:00	1X
Hold	4	∞	

10. Analyse 5 μ l of each RT-PCR reaction on a 1% agarose gel with DNA staining dye. When using Invitrogen E-Gel 48 Agarose precast gels, preload each well with 10 μ l of EB buffer. All bands are expected to be ~2.5kb, so run with an appropriate DNA ladder. It is preferable to run all set A and B reactions for one sample on the same row.

Part 4 - PCR pooling and cleanup

1. An example gel is provided below for two different samples:



2. Pool the individual amplicons for each sample into a single well of a clean plate or PCR strip. The aim here is to roughly balance the amount of DNA from each amplicon to provide even coverage across the genome.
3. Adjust the final volume of the pooled PCR product to 50 μ l by aliquoting when exceeding or adding an appropriate volume of clean EB buffer when less.
4. Add 40 μ l of room temperature AMPure XP beads (0.8X bead ratio). Gently pipette the entire volume up and down 15 times to mix thoroughly.
5. Incubate at room temperature for 10 min.
6. Place the plate/tubes onto the magnetic stand for at least 2 minutes, until the solution appears clear. Do not remove from magnetic stand during washing steps, and take care to not disturb beads.
7. Remove and discard the supernatant from each well by pipetting.
8. Add 200 μ l of freshly prepared 80% EtOH to each well without disturbing the beads.
9. Incubate at room temperature for 30 sec.
10. Remove and discard the supernatant from each well by pipetting.
11. Repeat steps 8 to 10 for a total of two EtOH washes.
12. Allow the beads to air dry for 5 to 15 min. Visually inspect wells to ensure any small droplets are removed by pipetting or evaporation.
13. Remove the plate/tubes from the magnetic stand.
14. Resuspend the dried bead pellet with 40 μ l of EB buffer.
15. Gently pipette the entire volume up and down 15 times to mix thoroughly.
16. Place the plate/tubes onto the magnetic stand at room temperature for at least 2 minutes, until the solution appears clear.
17. Transfer the cleared supernatant containing the purified DNA into a suitable plate, strip or tube. Use immediately or store frozen at -20°C .
18. Quantify all purified DNA using the Qubit dsDNA broad range assay.

19. For any samples >10 ng/μl, roughly dilute using EB buffer so the final concentration is between 1 - 10 ng /μl.
20. Re-quantify all purified and adjusted DNA using the Qubit dsDNA high sensitivity assay.
21. Dilute to 0.2 ng/μl using EB buffer (ensure at least 20 μl of volume) and proceed to library prep.

Part 5 - DNA library preparation and sequencing

The following protocol follows the Illumina Nextera XT DNA library prep kit except for two important changes: i) all reaction volumes are halved to save on reagents and ii) that following the clean up after library amplification, the libraries are manually normalised rather than with the provided normalisation beads.

1. Combine 5 μl Tagment DNA Buffer (TD) and 2.5 μl of pooled, diluted (0.2 ng/μl) amplicons for each sample into a PCR plate or strip.
2. Mix well by pipetting 10 times.
3. Add 2.5 μl Amplicon Tagment Mix (ATM) to each well on top of the TD/DNA mix.
4. Mix well by pipetting 10 times. Seal and briefly centrifuge.
5. Incubate the reaction mix as follows:

STEP	TEMP (°C)	TIME (mm:ss)
Tagmentation	55	05:00
Hold	10	∞

6. Following tagmentation, immediately remove the reaction from the thermocycler and add 2.5 μl Neutralize Tagment Buffer (NT) to stop the reaction.
7. Mix well by pipetting 10 times. Seal and briefly centrifuge.
8. Incubate at room temperature for 5 mins.
9. Add indices and Nextera PCR Master Mix (NPM) to the neutralised tagmentation reaction for each sample as below:

REAGENT	VOLUME (μl)
Neutralised tagmentation reaction	12.5
i7 adapter	2.5
i5 adapter	2.5
Nextera PCR Master Mix	7.5
TOTAL	25

10. Incubate the reaction as follows:

STEP	TEMP (°C)	TIME (mm:ss)	CYCLES
Hot start	72	03:00	1X
Initial denature	95	00:30	1X
Denaturation	95	00:10	12X
Annealing	55	00:30	
Extension	72	00:30	
Final extension	72	05:00	1X
Hold	10	∞	

11. Add 15 µl of room temperature AMPure XP beads (0.6X bead ratio) to the 25 µl of amplified libraries. Gently pipette the entire volume up and down 15 times to mix thoroughly.
12. Incubate at room temperature for 10 min.
13. Place the plate/tubes onto the magnetic stand for at least 2 minutes, until the solution appears clear. Do not remove from magnetic stand during washing steps, and take care to not disturb beads.
14. Remove and discard the supernatant from each well by pipetting.
15. Add 200 µl of freshly prepared 80% EtOH to each well without disturbing the beads.
16. Incubate at room temperature for 30 sec.
17. Remove and discard the supernatant from each well by pipetting.
18. Repeat steps 15 to 17 for a total of two EtOH washes.
19. Allow the beads to air dry for 5 to 15 min. Visually inspect wells to ensure any small droplets are removed by pipetting or evaporation.
20. Remove the plate/tubes from the magnetic stand.
21. Resuspend the dried bead pellet with 22.5 µl of EB buffer.
22. Gently pipette the entire volume up and down 15 times to mix thoroughly.
23. Place the plate/tubes onto the magnetic stand at room temperature for at least 2 minutes, until the solution appears clear.
24. Transfer 20 µl of the cleared supernatant containing the purified DNA into a suitable plate, strip or tube. Use immediately or store frozen at -20°C.
25. Quantify all purified DNA using the Qubit dsDNA high sensitivity assay.
26. Pool the individual libraries equally in DNA amount based on the Qubit values. Here, we assume the libraries will have similar fragment lengths and distributions.
27. Quantify the final pool of libraries using the Qubit dsDNA high sensitivity assay.
28. Analyse 2 µl of the final pool of libraries with Agilent High Sensitivity D5000 ScreenTape ensuring the whole fragment peak is captured.
29. Scale the calculated molarity from the TapeStation to the Qubit DNA concentration using the following formula:

$$\text{Final molarity} = (\text{TapeStation DNA molarity}) \times (\text{Qubit DNA conc} / \text{TapeStation DNA conc})$$

30. Dilute the final pooled libraries down 1 nM (at least 50 µl) using EB buffer, and add PhiX sequencing control if required.
31. Combine 20 µl of 1 nM library and 80 µl of EB buffer to dilute the final pool of libraries to 0.2 nM for loading.
32. Load 20 µl of libraries into a defrosted Illumina iSeq cartridge with flow cell, and sequence with at least 75 nt paired end sequencing. Assuming even pooling, each run can comfortably include 12-18 SARS-CoV-2 genome libraries with >1000x coverage for consensus calling.

Finally, assemble your viruses and upload them onto www.gisaid.org as soon as possible.